

Spring (2023)

compute coefficient of variation of following data :-
QUESTION NO 4

Classes	frequency	X	X ²	fx	fx ²
14-16	7	15	225	105	1575
17-19	9	18	324	162	2916
20-22	3	21	441	63	1323
23-25	8	24	576	192	4608
26-28	6	27	729	162	4374
29-31	4	30	900	120	3600
32-34	2	33	1089	66	2178

$$\Sigma f = 39 \quad \Sigma X = 168 \quad \Sigma X^2 = 4284 \quad \Sigma fx = 870 \quad \Sigma fx^2 = 20574$$

$$S = \sqrt{\frac{\Sigma fx^2}{\Sigma f} - \left(\frac{\Sigma fx}{\Sigma f}\right)^2}$$

$$S = \sqrt{\frac{20574}{39} - \left(\frac{870}{39}\right)^2}$$

$$S = \sqrt{527.538 - 497.633}$$

$$S = 5.4685$$

$$\bar{X} = \frac{870}{39}$$

$$\bar{X} = 22.31$$

Coefficient of Variation :-

$$C.V = \frac{S}{\bar{X}} \times 100$$

$$= \frac{5.4685}{22.31} \times 100 = 24.51\%$$

QUESTION NO 3 :-

Estimate the correlation coefficient between x and y

X	Y	XY	X ²	Y ²
45	110	4950	2025	12100
46	111	5106	2116	12321
47	111	5217	2209	12321
48	115	5520	2304	13225
49	116	5684	2401	13456
50	118	5900	2500	13924
51	120	6120	2601	14400
52	121	6292	2704	14641
53	124	6572	2809	15376
54	128	6912	2916	16384
$\Sigma X = 495$	$\Sigma Y = 1174$	$\Sigma XY = 58,273$	$\Sigma X^2 = 24,585$	$\Sigma Y^2 = 138,148$

$$S_{xy} = \frac{\Sigma XY}{n} - \left(\frac{\Sigma X}{n} \right) \left(\frac{\Sigma Y}{n} \right)$$

$$S_{xy} = \frac{58,273}{40} - \left(\frac{495}{10} \right) \left(\frac{1174}{10} \right)$$

$$S_{xy} = 5827.3 - (49.5)(117.4)$$

$$S_{xy} = 5827.3 - 5811.3$$

$$S_{xy} = 16$$

$$S_x = \sqrt{\frac{\Sigma X^2}{n} - \left(\frac{\Sigma X}{n} \right)^2}$$

$$S_x = \sqrt{\frac{24,585}{10} - \left(\frac{495}{10} \right)^2}$$

$$S_x = 2.87228$$

$$S_y = \sqrt{\frac{\Sigma Y^2}{n} - \left(\frac{\Sigma Y}{n} \right)^2}$$

$$S_y = \sqrt{\frac{138,148}{10} - \left(\frac{1174}{10} \right)^2}$$

$$S_y = 5.66038$$

Coefficient of Correlation :-

$$r = \frac{\sum xy}{\sum x \sum y}$$

$$r = \frac{16}{(2.87228)(5.6038)}$$

$$r = 0.984$$

Question no : 3 Spring 2022

X	Y	XY	X ²	Y ²
72	67	4824	5184	4489
54	57	3078	2916	3249
73	68	4964	5329	4624
63	64	4032	3969	4096
66	68	4488	4356	4624
65	72	4680	4225	5184
60	67	4020	3600	4489
70	73	5110	4900	5329
71	69	4899	5041	4761
69	72	4968	4761	5184
$\sum X =$	$\sum Y =$	$\sum XY =$	$\sum X^2 =$	$\sum Y^2 =$
663	677	45,063	44,281	46,029

$$S_{xy} = \frac{\sum XY}{n} - \left(\frac{\sum X}{n}\right)\left(\frac{\sum Y}{n}\right)$$

$$S_{xy} = \frac{45063}{10} - \left(\frac{663}{10}\right)\left(\frac{677}{10}\right)$$

$$S_{xy} = 4506.3 - (66.3)(67.7)$$

$$S_{xy} = 17.79$$

$$S_x = \sqrt{\frac{\sum X^2}{n} - \left(\frac{\sum X}{n}\right)^2} = \sqrt{\frac{44281}{10} - \left(\frac{663}{10}\right)^2}$$

$$S_x = \sqrt{32.41}$$

$$S_x = 5.69297$$

$$S_y = \sqrt{\frac{\sum Y^2}{n} - \left(\frac{\sum Y}{n}\right)^2} = \sqrt{\frac{46029}{10} - \left(\frac{677}{10}\right)^2}$$

$$S_y = \sqrt{19.61} = 4.428317$$

Coefficient of correlation

$$r = \frac{S_{xy}}{S_x S_y} = \frac{17.79}{(5.69297)(4.428317)}$$

$$r = 0.70566$$

Find regression line on Y on X

$$(Y - \bar{Y}) = b_{yx} (X - \bar{X})$$

$$b_{yx} = r \frac{S_y}{S_x}$$

$$Y - 67.7 = 0.5489 (X - 66.3)$$

$$Y - 67.7 = 0.5489 X - 36.39207$$

$$Y = 0.5489 X - 36.39207 + 67.7$$

$$Y = 0.5489 X + 31.30793$$

$$b_{yx} = 0.70566 \frac{4.428}{5.69}$$

$$b_{yx} = 0.5489$$

QUESTION NO 2 :-

Ages	frequency	X	fX	X ²	fX ²
14-15	6	14.5	87	210.25	1261.5
16-17	16	16.5	264	272.25	4356
18-19	20	18.5	370	342.25	6845
20-21	31	20.5	635.5	420.25	13027.75
22-23	15	22.5	337.5	506.25	7593.75
24-25	12	24.5	294	600.25	7203
	$\Sigma f = 100$	$\Sigma X = 117$	$\Sigma fX = 1988$	$\Sigma X^2 = 2351.3$	$\Sigma fX^2 = 40287$

Compute mean :-

$$\bar{X} = \frac{\Sigma fX}{\Sigma f} = \frac{1988}{100} = 19.88$$

Compute standard derivation :-

$$S = \sqrt{\frac{\Sigma fX^2 - (\Sigma fX)^2}{\Sigma f}}$$

$$S = \sqrt{\frac{40287 - \left(\frac{1988}{100}\right)^2}{100}}$$

$$S = 2.07668$$

Compute Coefficient of Variance

$$\begin{aligned} C.V &= \frac{S}{\bar{X}} \times 100 \\ &= \frac{2.07668}{19.88} \times 100 \\ &= 10.44\% \end{aligned}$$

Fall 2021 :- Past paper

Calculate the coefficient of correlation and regression line only on X for the following data?

X	Y	XY	X^2	Y^2
3	67	201	9	4489
4	57	228	16	3249
5	68	340	25	4624
6	64	384	36	4096
7	68	476	49	4624
8	72	576	64	5184
9	67	603	81	4489
10	73	730	100	5329
11	69	759	121	4761
12	72	864	144	5184
$\Sigma X = 75$		$\Sigma Y = 677$	$\Sigma XY = 5161$	$\Sigma X^2 = 645$
				$\Sigma Y^2 = 46029$

$$S_{XY} = \frac{\Sigma XY}{n} - \left(\frac{\Sigma X}{n}\right)\left(\frac{\Sigma Y}{n}\right)$$

$$S_{XY} = \frac{5161}{10} - \left(\frac{75}{10}\right)\left(\frac{677}{10}\right)$$

$$S_{XY} = 516.1 - (7.5)(67.7)$$

$$S_{XY} = 8.35$$

$$S_x = \sqrt{\frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2} = \sqrt{\frac{645}{10} - \left(\frac{75}{10}\right)^2}$$

$$S_x = 2.87228$$

$$S_y = \sqrt{\frac{\sum y^2}{n} - \left(\frac{\sum y}{n}\right)^2} = \sqrt{\frac{46,029}{10} - \left(\frac{677}{10}\right)^2}$$

$$S_y = 4.428317$$

$$r = \frac{S_{xy}}{S_x S_y} = \frac{8.35}{(2.87228)(4.428317)} = 0.65647$$

Find Regression line on Y on X

$$(Y - \bar{Y}) = b_{yx} (X - \bar{X})$$

$$Y - 67.7 = 1.01210 (X - 7.5)$$

$$Y = 1.01210 X - 7.59075 + 67.7$$

$$Y = 1.01210 X + 60.10925$$

$$b_{yx} = r \frac{S_y}{S_x}$$

$$b_{yx} = 0.65647 \frac{4.428317}{2.87228}$$

$$b_{yx} = 1.01210$$

QUESTION NO 2 :-

Compute mean, median, mode and coefficient of variation.

Weight	Frequency	X	fX	C.f	C.O.B	X ²	fX ²
118-126	3	122	366	3	117.5-126.5	14884	44652
127-135	5	131	655	8	126.5-135.5	17161	85805
136-144	9 f ₁	140	1260	17 ^c	135.5-144.5	19600	176400
145-153	12 f _m	149	1788	29	144.5-153.5	22201	266412
154-162	5 f ₂	158	790	34	153.5-162.5	24964	124820
163-171	4	167	668	38	162.5-171.5	27889	111556
172-180	2	176	352	40	171.5-180.5	30976	61952
	$\Sigma f = 40$	$\Sigma X = 1043$	$\Sigma fX = 5879$	$\Sigma = 40$			$\Sigma fX^2 = 871597$

Compute Mean :-

Model class

$$\bar{X} = \frac{\Sigma fX}{\Sigma f} = \frac{5879}{40} \quad \bar{X} = 146.975$$

Compute Median

$$\text{Median} = l + \frac{h}{f} \left(\frac{n}{2} - c \right)$$

$$\frac{n}{2} = \frac{\Sigma f}{2} = \frac{40}{2} = 20$$

$$\text{Median} = 144.5 + \frac{9}{12} (20 - 17)$$

$$\text{Median} = 144.5 + 0.75 (3)$$

$$\hat{X} = 146.75$$

Compute the mode :-

$$\hat{X} = l + \frac{f_m - f_1}{(f_m - f_1) + (f_m - f_2)} \times h$$

$$\hat{X} = 144.5 + \frac{12 - 9}{(12 - 9) + (12 - 5)} \times 9$$

$$\hat{X} = 144.5 + \frac{3}{10} \times 9$$

$$\hat{X} = 144.5 + 0.3 \times 90$$

$$\hat{X} = 147.2$$

Compute Coefficient of Variation :-

$$C.V = \frac{s}{\bar{X}} \times 100$$

$$s = \sqrt{\frac{\sum X^2}{n} - \left(\frac{\sum X}{n}\right)^2} = \sqrt{\frac{\sum fX^2}{\sum f} - \left(\frac{\sum fX}{\sum f}\right)^2}$$

$$s = \sqrt{\frac{871,597}{40} - \left(\frac{5879}{40}\right)^2}$$

$$s = \sqrt{21789.925 - 21601.68062}$$

$$s = 13.72$$

$$\bar{X} = \frac{\sum fX}{\sum f} = \frac{5879}{40} = 146.975$$

Coefficient of Variation :-

$$C.V = \frac{13.72}{146.975} \times 100$$

$$C.V = 0.0933 \times 100$$

$$C.V = 9.33 \%$$

Past Paper (2019)

QUESTION NO 5 :- Is the same spring 2022
question no 3

QUESTION NO 4 :- Is the same spring 2022
question no 2

Past paper (2018)

QUESTION NO 4 :- Is the same spring 2022
question no 2.

QUESTION NO 5 :- Is the same spring 2022
question no 3.

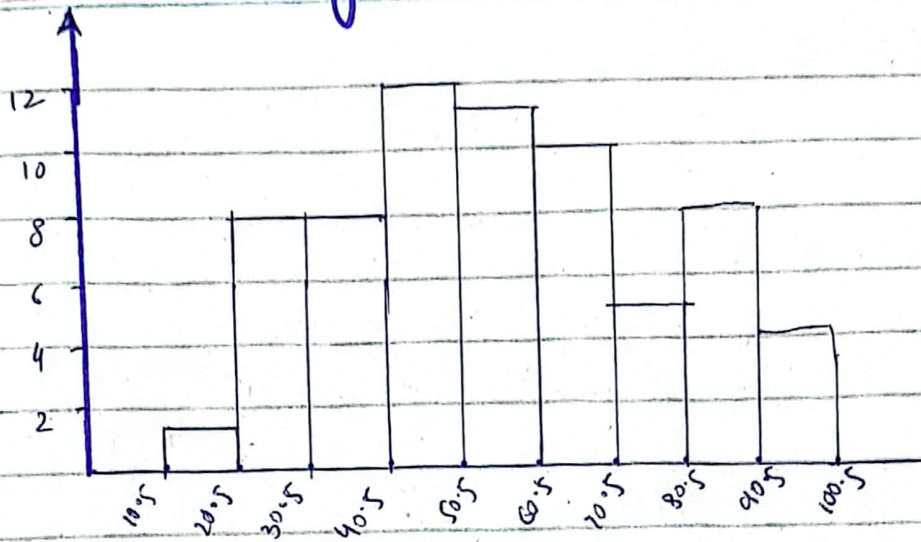
QUESTION NO 3 :-

Prepare a frequency table for the following
data taking classes (as 11-20, 21-30, ...).

Solution :

Class limit	Tally Marks	Class Boundaries	Frequency
11-20	I	10.5 - 20.5	1
21-30		20.5 - 30.5	8
31-40		30.5 - 40.5	8
41-50	 	40.5 - 50.5	12
51-60	 	50.5 - 60.5	11
61-70		60.5 - 70.5	10
71-80		70.5 - 80.5	5
81-90		80.5 - 90.5	8
91-100		90.5 - 100.5	3

Histogram

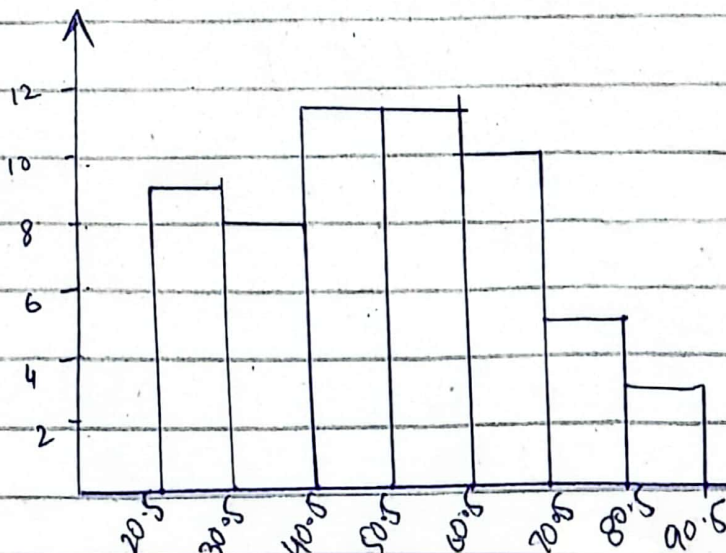


Past Paper 2017 :- STATISTICS

QUESTION NO 5

Solution:-

Class limit	frequency	Class boundaries	Tally mark
21-30	9	20.5 - 30.5	
31-40	8	30.5 - 40.5	
41-50	11	40.5 - 50.5	
51-60	11	50.5 - 60.5	
61-70	10	60.5 - 70.5	
71-80	5	70.5 - 80.5	
81-90	3	80.5 - 90.5	



QUESTION NO : 4

X	Y	XY	X ²	Y ²
65	57	3705	4225	3249
64	59	3776	4096	3481
73	68	4964	5329	4624
63	64	4032	3969	4096
66	68	4488	4356	4624
65	72	4680	4225	5184
60	67	4020	3600	4489
70	73	5110	4900	5329
71	69	4899	5041	4761
69	72	4968	4761	5184
$\Sigma X = 666$	$\Sigma Y = 669$	$\Sigma XY = 44642$	$\Sigma X^2 = 44502$	$\Sigma Y^2 = 45021$

$$S_{xy} = \frac{\Sigma XY}{n} - \left(\frac{\Sigma X}{n} \right) \left(\frac{\Sigma Y}{n} \right)$$

$$S_{xy} = \frac{44642}{10} - \left(\frac{666}{10} \right) \left(\frac{669}{10} \right)$$

$$S_{xy} = 4464.2 - (66.6)(66.9)$$

$$S_{xy} = 8.66$$

$$S_x = \sqrt{\frac{44502}{10} - \left(\frac{666}{10} \right)^2}$$

$$S_x = 3.8262$$

$$s_y = \sqrt{\frac{45021}{10} - \left(\frac{669}{10}\right)^2}$$

$$s_y = 5.1468$$

$$r = \frac{s_{xy}}{s_x s_y} = \frac{8.66}{(3.8262)(5.1468)}$$

$$r = 0.43915$$

Find regression line on Y on X

$$y - \bar{y} = b_{yx} (x - \bar{x})$$

$$b_{yx} = \frac{0.43915 \cdot 5.1468}{3.8262}$$

$$y - 66.9 = 0.59152 (x - 66.6)$$

$$b_{yx} = 0.59152$$

$$y - 66.9 = 0.59152x - 39.3952$$

$$y = 0.59152x - 39.3952 + 66.9$$

$$y = 0.59152x + 27.5048$$

QUESTION NO 3 :- Is same as past paper question no 2 (Spring 2022)